

(a)

word-final token emitted in chunk $k = \lfloor t_{end}/80 \text{ ms} \rfloor + \delta$ ($\delta = 2$ here)

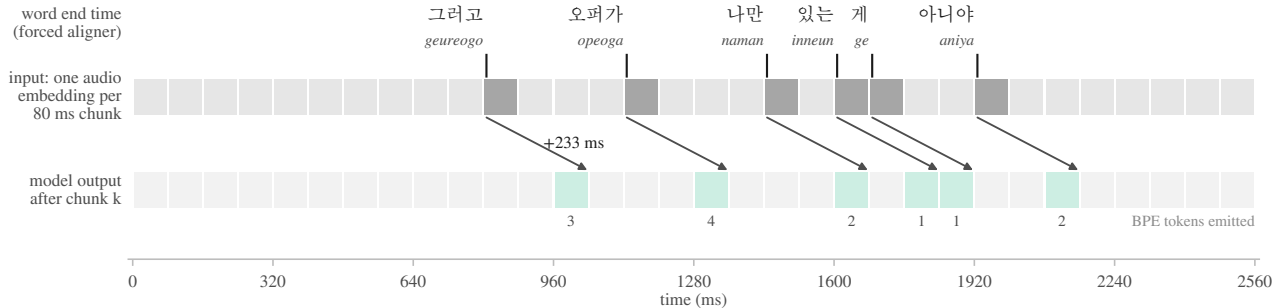
emission delay $= (k + 1) \cdot 80 \text{ ms} - t_{end}$: 160-240 ms for $\delta = 2$; 233 ms for each word below

audio chunk

chunk containing a word end

chunk ends with text tokens + <NEXT_AUDIO>

chunk ends with <NEXT_AUDIO> only

**(b)**

prefix (input only): ...assistant\nlanguage Korean<asr_text><DELAY_2>

chunks 11-13 ... [AUDIO_11] <NEXT_AUDIO> [AUDIO_12] 그 러 고 <NEXT_AUDIO> [AUDIO_13] <NEXT_AUDIO> ...

stream end ... [AUDIO_31] <NEXT_AUDIO> <EMPTY_AUDIO> <NEXT_AUDIO> flush round: nothing left to emit

— = next-token CE target: text tokens (weight 1) and <NEXT_AUDIO> (weight 0.3 EN / 0.15 KO)

no loss on prefix, [AUDIO_k] (adapter output replaces the audio placeholder embedding) and <EMPTY_AUDIO> positions